Physical Optics Sim - Documentation

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Purpose

This code implements fresnel and fraunhofer diffraction techniques as well as beam steering to accurately incorporate all aspects of the BICEP Array calibration process. Replicating this process can allow us to compare the data from the telescope beam maps to the simulations maps. Ultimatly finding a quantitative assessment of how the mirror non-flatness might be affecting areas of the beam maps.

Global Variables

Global variables allow for customazation of the simulation. The variables are as follows:

Variable	What it does
N	number of grid points
Radius	Radius of circular aperture
dx	spacing between grid points
edge taper	controls the gaussian fade
theta 1 and theta 2	Steering angles for tilt adjustments
lamda	wavelength of light being observed
mirror coord	Cartesian coordinates for the mirror system
window size	Size of the grid used in Fresnel diffraction
phase gradient	.
z far	Far-field distance in ()
k	wavenumber

Table 1: Available variables

N number approximated for optimalization via the equation below:

$$\Delta x \ll \frac{\lambda z}{2\text{Aperture Width}} \tag{1}$$

Giving us about a N number for pixels at 125.

Aperture Creation

The aperture function focuses on creating a 2d spatial grid that defines a circular aperture. It uses an N number of grid points along each axis, and grid spacing (dx) in the x and y directions. The aperture is gnerated with a Gaussian edge taper to simulate real teqniques. This means that some of the parts of the grid are non-zero, simulating a circular aperture. The function used to help create this is as follows:

$$r = \sqrt{x[i,j]^2 + y[i,j]^2}$$

The dx remains the same for the aperture function and the fresnel function, since both use the same physical grid spacing. (1)

The coordinate array is centered at zero and a gaussian taper is applied to replicate the illumination of the aperture in experimental values.

Beam steering teqniques are also applied during this step using a tilt phase calculation for the whole grid. This calculation can be seen below.

$$\phi = k(xcos(theta) + ysin(theta))tan(\alpha)$$

In order to steer the beam source two different angles must be defined. Here, there is a tilt angle (

$$\alpha$$

) and a rotation angle (

$$\theta$$

) on the x-axis. (insert ref)

alpha steer is the tilt angle. This describes the angle the beam makes in the z direction.

theta steer is the azamuth angle, which is the xy direction that the beam is leaning in.

spactial frequencies and slope can then be defined using trigonometry.

When both angles are zero, the tilt phase is zero, meaning that the beam is pointing straight ahead.

If you want to steer along the x or the y axis (right or left) the theta steer angle can be changed to achieve this.

This method can then be applied to different scenarios to achieve different directional steering.

Lastly, an adjustment is applied to ensure that the physics remains accurate.

$$\exp(1i * tilt_phase) \tag{2}$$

(insert ref)

Mirror Creation

The mirror function creates new coordinates that can be used for the fresnel calculation and later on in the fraunhofer calculation.

Since the mirror is non uniform, this fucntion helps to ensure that the fft used in the fraunhofer far field calculation is still accurate. This is where the lengths and widths of the codes are used to calculate a center for the mirror, along with ensuring that the dx or physical grid spacing is still accurate in both the x and y directions since one side will be longer than the other.

This function also focuses on applying the tilted coordinates forward along the y axis.

DISTORTION: When considering the non-flatness of the mirror, data points were used from (insert reference). I was able to reference code that was used in another paper (reference) to load in the photogrametry data points and bilinearly interpolate them into a digital representation. This representation can then be swapped into the simulation to replicate the actual mirror used in the calibration process.

Fresnel Diffraction

This function uses fresnel teqniques as found in (ref Optical physics third edition) to calculated the diffraction pattern from the aperture onto the mirror. Rather than using an FFT here, a double for loop is utilized with the diffraction by Huygen's principle:

$$E(x', y') = \iint A(x, y)\frac{e^{ikd}}{d}dxdy$$

This ensures that the actual math for the Fresnel diffraction is preserved and accurate in the near field, rather than approximating it with FFT teqniques.

This function also accounts for the complex phase shift that is created from the tilted mirror

SPEED FOR LOOPS One of the barriers for this kind of loop is how its processed.

This function applies a phase mesh to change the complex fields of the Near field output. The following phase array helps to convert uneven surface height on the mirror map into a phase delay map.

$$\phi(x, y) = \frac{2\pi}{\lambda} 2h(x, y) \tag{3}$$

This equation is broken down in the function to calculate the phase mask (derived from Eulers formula) and tranform the real numbers into complex 2d phasors. That are then matrix multiplied to give an output of the uneven mirror.

Fraunhofer Diffraction

A two step propogation: When doing Fraunhofer diffraction from the plane of the mirror to the far field, an FFT can be used.

Both a log scale and the linear scale of the FFTs are visable below:

The nearfield output from the fresnel diffraction function is input into the fraunhoffer function. This allows for all features applied to be carried through accurately into the far field projection.

Fresnel diffraction ingegral caluclation using a single FFT method explicity includes the distance and quadratic phase factor to output a grid in physical mm of distance. To go from this output to an angular deviation, the following equation can be used:

$$\tan(\theta_x) = \frac{x_2}{z_1}$$

Where z1 is the Far Field propogation distance. This formula yields an angle in radians which can then be converted to degrees for plot interpretation.

When considering the uneven axis on the mirror and how that's handled in the FFT we have to keep in mind the inverse relation with spatial dimensions.

$$f_x \propto \frac{1}{\delta x}$$

Since the mirror coordinates are stretched along the x-axis the FFT patern needs to look compressed on the x-axis.

Hygens principle for explination of variable c FOURIER SHIFT THEROEM:

Have to try beam tracking, to find out where the fft is projecting to, and then center the frame around that.

FIXING THE RADICAL TWO Y AXIS COMPRESSION

Optical propogation law:

$$Delta_{xout} = \frac{\lambda_z}{L_x} \tag{4}$$

Where L_x is equal to N_x times Δ_{xin}.

$$\Delta_{yout} = \frac{\lambda_z}{L_y} \tag{5}$$

Where L_y is equal to N_y times Δ_{yin}.

When the intermediate aperture of the near field mirror is used, the FFT calculation compresses the image in the y axis direction. This can be adjusted in the plotting steps when the y max and mins are calculated for the FFT plot. An adjustment factor of root two can help to recenter the image.

Normalization teqnuiques for graph comparision with difference between the flat and nonflat mirror fft.

Parseval's theorem:

This helps to ensure that even when different padding sizes are used, the total integrated energy still equals 1.0

Fitting to an eliptical 2D gaussian.

$$\delta(x) = \frac{1}{\omega} e^{-\frac{1}{2}(x-\mu)^T \epsilon (x-\mu)} \tag{6}$$

where epsilon equals the precision matrix:

$$\epsilon = \begin{bmatrix} r_x^2 & r_{xy}^2 \\ r_{xy}^2 & r_y^2 \end{bmatrix} \tag{7}$$

or using our precision matrix notation:

$$\epsilon = \begin{bmatrix} \sigma^2(1+P) & \sigma^2 C \\ \sigma^2 C & \sigma^2(1-P) \end{bmatrix} \tag{8}$$

This helped to tell us how far the nonflatness of the mirror spread or compressed the beam.

In this teqnique the normalized integral comparision is used to avoid losses in total energy concerning the beams shape.

Variable	What it does
xy	Frequency Coordinates
omega	Radius of circular aperture
mu_x	Mean peak (center) along the x-axis
mu_y	Mean peak along the y-axis
$r_x 2$	Precision matrix variable. Width along the x-axis
$r_y 2$	Precision matrix variable. Width along y axis.
$r_x y 2$	Precision matrix variable. Tilt or rotation of the eliptical beam.
ω	scaling factor

Table 2: Available variables for gaussian fit function

The max peak of the gaussian difference can help to indicate how severe the wavefront distortions from the non flat mirror are.

If the r_x and r_y values in the precision matrix are highly unequal this can indicate an elliptical beam. Given that the simulation is mostly ran with a 45 degree tilt on the mirror, this ellipticity should be expected.

REMOVING THE TILT BEFORE ANALYSIS:

Scalar diffraction theory

Coordinate rotation matrix to fix FFT stretching in gaussian elliptical Need to rotate the plane in the frequency domain, due to FFT Created from tilt in mirror plane. A matrix like the one below can be considered:

$$\begin{bmatrix} E_{xx} & E_{xy} \\ E_{xy} & E_{yy} \end{bmatrix} \tag{9}$$

Where the variables mean the following:

$$E_{xx} = \sigma_x^2 cos^2(\theta) + \sigma_y^2 sin^2(\theta) \tag{10}$$

$$E_{yy} = \sigma_x^2 sin^2(\theta) + \sigma_y^2 cos^2(\theta) \tag{11}$$

$$E_{xy} = (\sigma_x^2 - \sigma_y^2) sin(\theta) cos(\theta) \tag{12}$$

$fy_p rjected$ helps to counterbalance the inverse spatial compression alighning the covariance matrix back into circular dimensions.

Table 3: Far-Field Comparison of Original vs. New Near-Field Surfaces

Beam Parameter	Original Surface	New Surface	Physical Impact
Peak Amplitude ($1/\omega$)	0.47528	0.45761	-3.7% Power attenuation
Center Shift X (μ_x)	-0.0000	-0.2045	Subtle horizontal steering
Center Shift Y (μ_y)	-0.0001	-0.4960	Dominant vertical steering
True Width r_x^2	1.9000	1.9000	No horizontal defocusing
True Width r_y^2	1.9127	1.9666	+2.8% Beam stretching
Beam Tilt Angle (θ)	45.00°	45.00°	Aligned to projection axis
Covariance Term (xy)	0.0077	0.0313	Structural asymmetry increase

$$\text{exponent} = -\frac{1}{2}\mathbf{x}^T P \mathbf{x} = -\frac{1}{2}\begin{bmatrix} dx & dy \end{bmatrix}\begin{bmatrix} P_{xx} & P_{xy} \\ P_{xy} & P_{yy} \end{bmatrix}\begin{bmatrix} dx \\ dy \end{bmatrix}$$

$$\text{exponent} = -\left(\frac{P_{xx}}{2}dx^2 + P_{xy}dxdy + \frac{P_{yy}}{2}dy^2\right)$$

$$\text{exponent} = -(a \cdot dx^2 + 2b \cdot dxdy + c \cdot dy^2)$$

$$\begin{aligned} \frac{P_{xx}}{2} = a \quad &\Rightarrow \quad P_{xx} = 2a \\ P_{xy} = 2b \quad &\Rightarrow \quad P_{xy} = 2b \\ \frac{P_{yy}}{2} = c \quad &\Rightarrow \quad P_{yy} = 2c \end{aligned}$$

To compute the inverse of the Precision Matrix:

$$\det(P) = (P_{xx} \cdot P_{yy}) - (P_{xy} \cdot P_{xy})$$

Substituting:

$$\det(P) = (2a \cdot 2c) - (2b \cdot 2b) = 4ac - 4b^2 = \mathbf{4(ac - b^2)}$$

$$\Sigma = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix} = \frac{1}{\det(P)}\begin{bmatrix} P_{yy} & -P_{xy} \\ -P_{xy} & P_{xx} \end{bmatrix}$$

$$\Sigma = \frac{1}{4(ac - b^2)}\begin{bmatrix} 2c & -2b \\ -2b & 2a \end{bmatrix}$$

$$\sigma_{xx} = \frac{2c}{4(ac - b^2)} = \frac{\mathbf{2c}}{\det}$$

$$\sigma_{yy} = \frac{2a}{4(ac - b^2)} = \frac{\mathbf{2a}}{\det}$$

$$\sigma_{xy} = \frac{-2b}{4(ac - b^2)} = \frac{\mathbf{-2b}}{\det}$$

References

Phase adjustment for non-flat mirror and phase array equation:

Peak Normalization - Fraunhofer function: https://blogs.bath.ac.uk/python/finding-maximum-values-from-arrays/

Diffraction Scaling - Voelz, David George, and Society of Photo-optical Instrumentation Engineers. Computational Fourier Optics a MATLAB Tutorial. SPIE, 2011, https://doi.org/10.1117/3.858456.

Goodman, Joseph W. Introduction to Fourier Optics. McGraw-Hill, 1968.